

Rigorous interval certificates for selected windows

PRIME.RDAGGER.INTERVAL.CERT.01 — round 465

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Abstract

The production-faithful selected windows of round 463 are certified by outward interval arithmetic at every fixed index $k = 5, \dots, 12$. The proof encloses the complete von-Mangoldt source, folded signed moments, two nested moment matrices, the smooth-border functional, and the terminal quotient $q^\dagger$. A naive interval Stieltjes–Lanczos recurrence is complete only for $k = 5, 6$; dependency widening makes its norm interval contain zero at degrees 11, 11, 10, 10, 9, 8 for $k = 7, \dots, 12$. The stable route is therefore a direct Chebyshev moment certificate with an exact basis change, validated Cholesky lower bounds, and residual quadratic-form enclosures.

The resulting theorem is finite:

$$k \in \{5, \dots, 12\} \implies R^\dagger(W_k) \succeq \tfrac{1}{2}I.$$

It does not prove the frequently-selected quantifier, close any external Lean dictionary, or prove the Riemann hypothesis.

Status. **CERTIFIED** for the eight sealed finite computational objects. No promotion of an RH claim. No claim for or against RH.

1 Exact sealed object

For $k \geq 5$ put

$$\begin{aligned} r_k &= \lfloor \sqrt{k} \rfloor, & a_k &= 2^k, & m_k &= k2^{r_k} - 1, & M_k &= m_k + 1, \\ L_k &= 2M_k - 2, & N_k &= M_k/2, & \Delta_k &= \frac{\log a_k}{M_k} = 2^{-r_k} \log 2. \end{aligned}$$

This is the selected geometry used in `RH/FaithfulFold.lean`. The complete arithmetic lag is

$$c_i^P = - \sum_{\substack{n=p^\ell \leq a_k^2 \\ p \text{ prime}}} \frac{\log p}{\sqrt{n}} [t_{\Delta_k}(i\Delta_k - \log n) + t_{\Delta_k}(i\Delta_k + \log n)], \quad t_\Delta(x) = \max(0, 1 - |x|/\Delta).$$

The prime-power list is generated by an integer Eratosthenes sieve; n , p , a_k , m_k , M_k , L_k , and all tent coefficients are exact integers or rationals. The operations $\log p$, $\log n$, $\sqrt{n}$, and $\log 2$ are evaluated by outward `mpmath.iv` intervals. For $k \geq 5$ one has $\Delta_k \leq \log 2/4$, so the reflected prime tent is rigorously zero. Moreover $n \geq a_k$ implies $\log n \geq M_k \Delta_k$, hence all its tents at $i = 0, \dots, M_k - 1$ vanish. The implementation still counts every prime power through a_k^2 , but evaluates only the proved-active prefix $n < a_k$.

The archimedean lag c_i^A is the `r463 productionArchLag`: the deployed 48-node Gauss–Legendre realization of the two smooth segments. Each Legendre root is enclosed in radius 10^{-90} at 120 decimal digits; opposite endpoint signs of P_{48} , disjoint root intervals, and $2 \in \sum_{\ell=1}^{48} \omega_\ell$ are checked. Transcendental evaluation uses `GL.N=48` and `DPS=70`. In the zero-lag branch the constants are outward enclosures of Euler’s constant and $\log \pi$.

The smooth completion is the production grid

$$u_\ell = \frac{\ell}{100}, \quad 1 \leq \ell, \quad u_\ell < 2 \log a_k, \quad \beta_\ell = \frac{2}{100} e^{u_\ell/2},$$

sent through the same ordinary tents, archimedean lag, spectral density, and fold. This fixes the r463 completion data used here; its budget is not an unrelated opaque scalar.

To include the literal deployed r463 arrays as well as the ideal interval realization, each lag interval is finally hulled with the corresponding binary64 production value. A binary64 endpoint is an exact dyadic rational. It is not trusted as an approximation: enlarging an interval can only make certification harder. This compatibility hull explains the visible widths at $k \geq 9$ and ensures that every r459/r463 point value lies inside the published certificate.

For $1 \leq j \leq M_k - 1$ set

$$\begin{aligned} \theta_j &= \frac{2\pi j}{L_k}, & x_j &= \cos \theta_j, \\ d_j &= c_0 + 2 \sum_{i=1}^{M_k-2} c_i \cos(i\theta_j) + c_{M_k-1} \cos((M_k-1)\theta_j), \\ w_j &= \frac{2}{L_k} (1 - \cos \theta_j) d_j, \end{aligned}$$

with w_{M_k-1} divided by two. These are exactly the r463 lag $\rightarrow$ spectral density $\rightarrow L = 2M - 2$ fold rules. The positive and negative parts of w_j are the two production channels. The interval signs are fixed on all eight windows.

2 Stable certificate route

Let $\sigma_k = \sum_j w_j \delta_{x_j}$ be the main signed fold and let τ_k be the signed smooth-border fold. In the Chebyshev basis $T_i(\cos \theta) = \cos(i\theta)$ define, for $n \leq N_k$,

$$(H_n)_{ab} = \int T_a T_b d\sigma_k, \quad (g_n)_a = \int T_a d\tau_k, \quad 0 \leq a, b < n.$$

No sign-split subtraction is performed numerically: the signed weights enter directly. Product-to-sum gives

$$(H_n)_{ab} = \frac{h_{|a-b|} + h_{a+b}}{2}, \quad h_s = \int T_s d\sigma_k,$$

which avoids both Vandermonde powers and the interval dependency of a Stieltjes recurrence.

For each $n = N_k - 1, N_k$, a frozen binary64 inverse-Cholesky factor is used as a fixed basis matrix C_n . Its entries are interpreted as exact dyadic rationals and its diagonal is nonzero. Thus $C_n^T H_n C_n \succ 0$ is equivalent to $H_n \succ 0$; no computed eigenvalue enters the theorem. On the transformed interval matrix the checker proves

$$\lambda_{\min}(C_n^T H_n C_n) \geq \mu_n > 0$$

by Cholesky of the midpoint shifted by half the floating hint, Higham's backward-error bound

$$|\delta A| \leq \gamma_{n+1} |L| |L^T|, \quad \gamma_s = \frac{(s+2)2^{-53}}{1 - (s+2)2^{-53}},$$

and subtraction of the interval-radius row sum. Every radius is inflated by $1 + 2^{-38}$ and 10^{-300} .

Lemma 1 (Residual enclosure). *If $H \succeq \mu I$ with $\mu > 0$, then for every seed x and $r = g - Hx$,*

$$g^T H^{-1} g = 2g^T x - x^T H x + r^T H^{-1} r \in \left[2g^T x - x^T H x, 2g^T x - x^T H x + \frac{\|r\|_2^2}{\mu} \right].$$

The displayed endpoints are interpreted as outward intervals when H and g are interval data.

Proof. Write $g = Hx + r$ and expand both sides. Since $0 \preceq H^{-1} \preceq \mu^{-1}I$, the remainder lies in $[0, \|r\|_2^2/\mu]$. $\square$

The binary64 solve is only a seed for Lemma 1; a poor seed widens the enclosure and cannot create a pass. Set

$$s_n = g_n^T H_n^{-1} g_n.$$

The r453 partial-Parseval identity and the production completion give

$$B_w = s_{N_k-1} + \frac{5}{7}, \quad q_k^\dagger = \frac{s_{N_k}}{B_w}.$$

Independent lower/upper residual enclosures of s_{N_k-1} and s_{N_k} therefore yield

$$\frac{\underline{s}_{N_k}}{\bar{s}_{N_k-1} + 5/7} \leq q_k^\dagger \leq \frac{\bar{s}_{N_k}}{\underline{s}_{N_k-1} + 5/7}.$$

Theorem 1 (Finite selected-window certificate). *For the exact production-faithful objects above,*

$for\ every\ k \in \{5, \dots, 12\}, \quad R^\dagger(W_k) \succeq \frac{1}{2}I$

is rigorously certified.

Proof. The checker proves $H_{N_k-1} \succ 0$ and $H_{N_k} \succ 0$ by the validated Cholesky bounds just described. It then applies Lemma 1; Table 1 gives outward intervals strictly contained in $(0, 1)$ for every $q_k^\dagger$. The r460 Schur-complement identity identifies $1 - q_k^\dagger$ as the last pivot of the augmented dual-resolvent form. Its leading block is positive by $H_{N_k} \succ 0$, so the augmented form is positive definite. The production whitening congruence then gives $R^\dagger(W_k) - \frac{1}{2}I \succeq 0$. $\square$

3 Certificate table and scaling

k	atoms	S	cap	$q^\dagger$ enclosure	width	seconds	chain
5	198	19	10	[.877898027316, .877898027326]	$1.000 \cdot 10^{-11}$	.200	10/10
6	604	23	12	[.885355701542, .885355701573]	$3.100 \cdot 10^{-11}$	.253	12/12
7	1961	27	14	[.860490813792, .860490813910]	$1.180 \cdot 10^{-10}$	.312	11/14
8	6635	31	16	[.852160102837, .852160103131]	$2.940 \cdot 10^{-10}$	.386	11/16
9	23150	71	36	[.8975761165, .8975761982]	$8.170 \cdot 10^{-8}$	1.610	10/36
10	82267	79	40	[.9319617974, .9319619463]	$1.489 \cdot 10^{-7}$	2.032	10/40
11	296347	87	44	[.9201637350, .9201641094]	$3.744 \cdot 10^{-7}$	2.559	9/44
12	1078555	95	48	[.9139227630, .9139233990]	$6.360 \cdot 10^{-7}$	3.159	8/48

Table 1: Sealed outward decimal certificates. “chain” is the last degree reached by naive interval Stieltjes–Lanczos before its norm interval contains zero. Timings exclude the common sieve setup.

The full run, including the common sieve through 4^{12} , takes 11.170 seconds and passes 14/14 gates. The same-plateau cases $k = 5, \dots, 8$ scale gently. At $k = 9$, M jumps from 32 to 72, and the binary64 compatibility hull is amplified by the deeper moment solve; the width jumps from $2.94 \cdot 10^{-10}$ to $8.17 \cdot 10^{-8}$. From $k = 9$ through 12, the direct matrix work grows approximately cubically in cap but remains below four seconds per member. The present complete-prime-power list grows like $4^k / \log(4^k)$ even though only $n < a_k$ is active. Thus $k = 13$ is expected to cost roughly four times the sieve memory and list traffic; $k = 14$ would require a streaming count/sieve redesign rather than a larger interval precision. This is an implementation scaling limit, not a certificate failure.

4 Why the chain was abandoned

The same lag hull was propagated through a naive monic interval Stieltjes recurrence. It certifies all 10 and 12 required degrees at $k = 5, 6$, but reaches only

$$11/14, 11/16, 10/36, 10/40, 9/44, 8/48$$

for $k = 7, \dots, 12$. At the next degree the interval norm contains zero, so division would be invalid. Increasing transcendental precision does not remove the deployed binary64 hull; recurrence dependency repeatedly reuses the same uncertain coefficients and widens algebraically. The machine therefore reports refusal and switches to the direct moment matrix. It never treats the midpoint chain as a certificate.

5 Lean interoperability design

There are two honest ports.

Exact dyadic data block. Export each sealed decimal endpoint, the binary64 basis matrix, the shift, Cholesky factor, and row-sum allowance as integers over powers of two or ten. A small Lean checker can reconstruct the rational matrix inequalities and decide positive pivots using `native_decide`. Transcendental source enclosures remain separate verified-data hypotheses, with hashes tying them to the Python generator.

Direct rational proof terms. Round every interval endpoint outward to a rational and export fraction-free LDL^T witnesses for the shifted transformed moment matrices. Lean can verify reconstruction and positive pivots with `norm_num`; the residual norm and quotient inequalities are then rational arithmetic. This is slower to generate but minimizes trusted executable code.

Either port should expose a proposition such as

$$\text{SelectedFiniteCert } k : 5 \leq k \wedge k \leq 12 \longrightarrow R^\dagger(W_k) \succeq \tfrac{1}{2}I.$$

It does not prove the frequently quantifier $\exists^\infty k, R^\dagger(W_k) \succeq \tfrac{1}{2}I$. Finite certificates formalize base cases and test the object dictionary; they cannot supply the missing cofinal mechanism. Moreover, r463 still presents `productionArchLag` and `productionCompletion` through interfaces. A Lean data port must identify the exported blocks with those interfaces; otherwise it proves only a parallel rational object.

6 Kill and gate table

Gate	result	evidence
Exact source and geometry	CERTIFIED	Integer prime-power sieve through a_k^2 ; rational mesh and smooth grid; outward log, exp, sqrt, pi, cosine.
Gauss–Legendre enclosure	CERTIFIED	48 root sign changes, disjoint intervals, and weight sum enclosing 2; root radius 10^{-90} .
r463 dimensions and fold	CERTIFIED	Atoms/ S /cap pins include 198/19/10, 23150/71/36, and 82267/79/40; endpoint half weight retained.
r459/r463 consistency	CERTIFIED	All eight reference values lie in the published intervals.
Determinism	CERTIFIED	Live outward bounds must lie inside byte-pinned decimal hulls; smoke and full use the same constants.
Stieltjes interval propagation	REFUSED-WIDTH	Complete at $k = 5, 6$; zero-containing norm interval at the documented degree for every $k \geq 7$.
Direct moment/Cholesky route	CERTIFIED	Validated lower bounds remain positive for both nested blocks; residual remainder bounded by $\ r\ ^2/\mu$.
Runtime	CERTIFIED	Smoke 0.349 s; full 11.170 s, below 45 minutes.
Infinite/frequently conclusion	not claimed	Eight finite cases do not imply an infinite subsequence.
Analytic/Lean dictionary	not claimed	The exported completion still requires identification with the opaque Lean interfaces and the external Weil bridges.

Claim boundary

This is a rigorous theorem about eight fixed, explicitly sealed production-faithful finite matrices. It does not prove the frequently selected statement, does not close the source-derived cofinal spectral-overlap gap of r460, and does not validate the external explicit-formula or density bridges. No RH claim and no anti-RH claim are made.